

ASSIGNMENT – 3.3

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Batch: 05

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Task 1: AI-Generated Logic for Reading Consumer Details

Scenario

An electricity billing system must collect accurate consumer data.

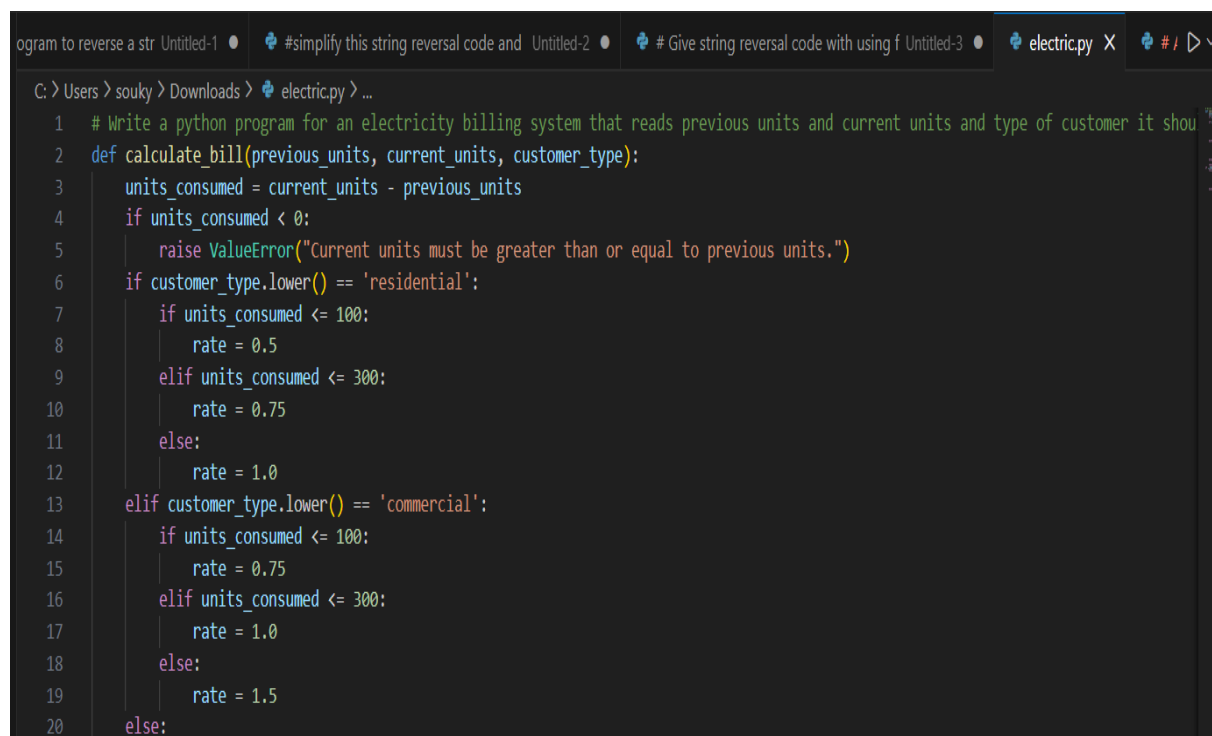
Task Description

Use an AI tool (GitHub Copilot / Gemini) to generate a Python program that:

- Reads:
 - o Previous Units (PU)
 - o Current Units (CU)
 - o Type of Customer
- Calculates units consumed
- Implements logic directly in the main program (no functions)

Expected Output

- Correct input reading
- Units consumed calculation
- Screenshot showing AI-generated code
- Sample input and output



```
C:\Users> souky > Downloads > electric.py > ...
1  # Write a python program for an electricity billing system that reads previous units and current units and type of customer it should
2  def calculate_bill(previous_units, current_units, customer_type):
3      units_consumed = current_units - previous_units
4      if units_consumed < 0:
5          raise ValueError("Current units must be greater than or equal to previous units.")
6      if customer_type.lower() == 'residential':
7          if units_consumed <= 100:
8              rate = 0.5
9          elif units_consumed <= 300:
10             rate = 0.75
11         else:
12             rate = 1.0
13     elif customer_type.lower() == 'commercial':
14         if units_consumed <= 100:
15             rate = 0.75
16         elif units_consumed <= 300:
17             rate = 1.0
18         else:
19             rate = 1.5
20     else:
```

```
2 def calculate_bill(previous_units, current_units, customer_type):
13     elif customer_type.lower() == "commercial":
14         if units_consumed <= 100:
15             rate = 0.75
16         elif units_consumed <= 300:
17             rate = 1.0
18         else:
19             rate = 1.5
20     else:
21         raise ValueError("Invalid customer type. Please enter 'residential' or 'commercial'.")
22     return units_consumed, units_consumed * rate
23 if __name__ == "__main__":
24     try:
25         previous_units = float(input("Enter previous meter reading (units): "))
26         current_units = float(input("Enter current meter reading (units): "))
27         customer_type = input("Enter customer type (residential/commercial): ")
28         units_consumed, total_bill = calculate_bill(previous_units, current_units, customer_type)
29         print(f"Units Consumed: {units_consumed}")
30         print(f"Total Bill Amount: ${total_bill:.2f}")
31     except ValueError as e:
32         print(f"Error: {e}")
```

PROBLEMS 18 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/souky/Downloads/electric.
Enter previous meter reading (units): 5000
Enter current meter reading (units): 6500
Enter customer type (residential/commercial): residential
Units Consumed: 1500.0
Total Bill Amount: $1500.00
PS C:\Users\souky>
```

Task 2: Energy Charges Calculation Based on Units Consumed

Scenario

Energy charges depend on the number of units consumed and customer type.

Task Description

Review the AI-generated code from Task 1 and extend it to:

- Calculate Energy Charges (EC)
- Use conditional statements based on:
 - o Domestic
 - o Commercial
 - o Industrial consumers
- Improve readability using AI prompts such as:
 - o "Simplify energy charge calculation logic"
 - o "Optimize conditional statements"

Expected Output

- Correct EC calculation
- Clear conditional logic
- Original and improved versions

```
C:\> Users > souky > Downloads > Untitled-6.py > ...
1 #In above code energy charges depend on number of units consumed and customer type .now it should calculate energy charge on domes
2 def calculate_bill(previous_reading, current_reading, customer_type):
3     units_consumed = current_reading - previous_reading
4     if units_consumed < 0:
5         raise ValueError("Current unit reading must be greater than or equal to previous unit reading.")
6     if customer_type.lower() == "residential":
7         if units_consumed <= 100:
8             rate = 0.5
9         elif units_consumed <= 300:
10            rate = 0.75
11        else:
12            rate = 1.0
13    elif customer_type.lower() == "commercial":
14        if units_consumed <= 100:
15            rate = 0.75
16        elif units_consumed <= 300:
17            rate = 1.0
18        else:
19            rate = 1.5
20    elif customer_type.lower() == "industrial":
21        if units_consumed <= 100:
22            rate = 1.0
```

```
... ring reversal code and Untitled-2 • # Give string reversal code with using f Untitled-3 • electric.py • Untitled-6.py X # Analyze the code with functions a
C:\> Users > souky > Downloads > Untitled-6.py > ...
2 def calculate_bill(previous_reading, current_reading, customer_type):
18         else:
19             rate = 1.5
20     elif customer_type.lower() == "industrial":
21         if units_consumed <= 100:
22             rate = 1.0
23         elif units_consumed <= 300:
24             rate = 1.5
25         else:
26             rate = 2.0
27     else:
28         raise ValueError("Invalid customer type")
29     return units_consumed * rate
30 if __name__ == "__main__":
31     try:
32         previous_units = int(input("Enter previous unit reading: "))
33         current_units = int(input("Enter current unit reading: "))
34         customer_type = input("Enter customer type (residential/commercial/industrial): ")
35         total_bill = calculate_bill(previous_units, current_units, customer_type)
36         print(f"Total bill amount: ${total_bill:.2f}")
37     except ValueError as e:
38         print(f"Error: {e}")
39
PROBLEMS 10 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/souky/Downloads/electric.py
Total Bill Amount: $1500.00
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/souky/Downloads/Untitled-6.py
File "c:\Users\souky\Downloads\Untitled-6.py", line 37
SyntaxError: expected 'except' or 'finally' block
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/souky/Downloads/Untitled-6.py
Enter previous unit reading: 5000
Enter current unit reading: 6500
Enter customer type (residential/commercial/industrial): commercial
Total bill amount: $2250.00
PS C:\Users\souky>
```

Task 3: Modular Design Using AI Assistance (Using Functions)

Scenario

Billing logic must be reusable for multiple consumers.

Task Description

Use AI assistance to generate a Python program that:

- Uses user-defined functions to:
 - o Calculate Energy Charges
 - o Calculate Fixed Charges
- Returns calculated values
- Includes meaningful comments

Expected Output

- Function-based Python program
- Correct EC and FC values
- Screenshots of AI-assisted function generation
- Test cases with outputs

```
C:\Users> souky > Downloads > #In above code billing logic must be reusable > ...
1 #In above code billing logic must be reusable for multiple consumers it should use defined functions to calculate energy charges, f
2 def calculate_energy_charges(units_consumed, customer_type):
3     if customer_type == 'residential':
4         if units_consumed <= 100:
5             rate = 0.5
6         elif units_consumed <= 300:
7             rate = 0.75
8         else:
9             rate = 1.0
10    elif customer_type == 'commercial':
11        if units_consumed <= 100:
12            rate = 0.75
13        elif units_consumed <= 300:
14            rate = 1.0
15        else:
16            rate = 1.5
17    elif customer_type == 'industrial':
18        if units_consumed <= 100:
19            rate = 1.0
20        elif units_consumed <= 300:
21            rate = 1.5
```

```
C:\Users> souky > Downloads > #In above code billing logic must be reusable > ...
20 elif units_consumed <= 300:
21     rate = 1.5
22 else:
23     rate = 2.0
24 else:
25     raise ValueError("Invalid customer type")
26 return units_consumed * rate
27 def calculate_fixed_charges(customer_type):
28     if customer_type == 'residential':
29         return 50
30     elif customer_type == 'commercial':
31         return 100
32     elif customer_type == 'industrial':
33         return 150
34     else:
35         raise ValueError("Invalid customer type")
36 def calculate_bill(previous_units, current_units, customer_type):
37     units_consumed = current_units - previous_units
38     if units_consumed < 0:
39         raise ValueError("Current unit reading must be greater than or equal to previous unit reading")
40     energy_charges = calculate_energy_charges(units_consumed, customer_type)
41     fixed_charges = calculate_fixed_charges(customer_type)
42     return energy_charges + fixed_charges
```

```
C:\Users> souky > Downloads > #In above code billing logic must be reusable > ...
36 def calculate_bill(previous_units, current_units, customer_type):
37     units_consumed = current_units - previous_units
38     if units_consumed < 0:
39         raise ValueError("Current unit reading must be greater than or equal to previous unit reading")
40     energy_charges = calculate_energy_charges(units_consumed, customer_type)
41     fixed_charges = calculate_fixed_charges(customer_type)
42     return energy_charges + fixed_charges
43 if __name__ == "__main__":
44     try:
45         previous_units = int(input("Enter previous unit reading: "))
46         current_units = int(input("Enter current unit reading: "))
47         customer_type = input("Enter customer type (residential/commercial/industrial): ")
48         bill_amount = calculate_bill(previous_units, current_units, customer_type)
49         print(f"Total bill amount: ${bill_amount:.2f}")
50     except ValueError as e:
51         print(f"Error: {e}")
```

```
ust be reu.py"
except ValueError as e:
    ^^^^^^
SyntaxError: invalid syntax
● PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/py
ust be reu.py"
Enter previous unit reading: 5000
Enter current unit reading: 6500
Enter customer type (residential/commercial/industrial): residential
Total bill amount: $1550.00
○ PS C:\Users\souky> |
```

Task 4: Calculation of Additional Charges

Scenario

Electricity bills include multiple additional charges.

Task Description

Extend the program to calculate:

- FC – Fixed Charges
- CC – Customer Charges
- ED – Electricity Duty (percentage of EC)

Use AI prompts like:

- “Add electricity duty calculation”
- “Improve billing accuracy”

Expected Output

- Individual charge values printed
- Correct duty calculation
- Well-structured output
- Verified intermediate results

```
using f Untitled-3 • electric.py • Untitled-6.py • #In above code billing logic must be reu.py • # In above code it should include multip.py • # A >
C: > Users > souky > Downloads > # In above code it should include multip.py > ...
1 # In above code it should include multiple additional charges liked fixedcharge , customercharge , electicityduty
2 def calculate_energy_charges(units_consumed,customer_type):
3     if customer_type == 'residential':
4         if units_consumed <= 100:
5             rate = 0.5
6         elif units_consumed <= 300:
7             rate = 0.75
8         else:
9             rate = 1.0
10    elif customer_type == 'commercial':
11        if units_consumed <= 100:
12            rate = 0.75
13        elif units_consumed <= 300:
14            rate = 1.0
15        else:
16            rate = 1.5
17    elif customer_type == 'industrial':
18        if units_consumed <= 100:
19            rate = 1.0
20        elif units_consumed <= 300:
21            rate = 1.5
22        else:
23            rate = 2.0
```

```
C: > Users > souky > Downloads > # In above code it should include multip.py > ...
21 def calculate_energy_charges(units_consumed, customer_type):
22     rate = 1.5
23     else:
24         rate = 2.0
25     else:
26         raise ValueError("Invalid customer type")
27     return units_consumed * rate
28 def calculate_fixed_charge(customer_type):
29     if customer_type == 'residential':
30         return 50.0
31     elif customer_type == 'commercial':
32         return 100.0
33     elif customer_type == 'industrial':
34         return 150.0
35     else:
36         raise ValueError("Invalid customer type")
37 def calculate_customer_charge(customer_type):
38     if customer_type == 'residential':
39         return 20.0
40     elif customer_type == 'commercial':
41         return 40.0
42     elif customer_type == 'industrial':
43         return 60.0
```

```
C: > Users > souky > Downloads > # In above code it should include multip.py > ...
36 def calculate_customer_charge(customer_type):
41     elif customer_type == 'industrial':
42         return 60.0
43     else:
44         raise ValueError("Invalid customer type")
45 def calculate_electricity_duty(energy_charges):
46     return energy_charges * 0.05 # 5% electricity duty
47 def calculate_bill(previous_units, current_units, customer_type):
48     units_consumed = current_units - previous_units
49     if units_consumed < 0:
50         raise ValueError("Current unit reading must be greater than or equal to previous unit reading")
51     energy_charges = calculate_energy_charges(units_consumed, customer_type)
52     fixed_charge = calculate_fixed_charge(customer_type)
53     customer_charge = calculate_customer_charge(customer_type)
54     electricity_duty = calculate_electricity_duty(energy_charges)
55     subtotal = energy_charges + fixed_charge + customer_charge
56     electricity_duty = calculate_electricity_duty(subtotal)
57     return subtotal + electricity_duty
58 if __name__ == "__main__":
59     try:
60         previous_units = int(input("Enter previous unit reading: "))
61         current_units = int(input("Enter current unit reading: "))
62         customer_type = input("Enter customer type (residential/commercial/industrial): ")
63         bill = calculate_bill(previous_units, current_units, customer_type)
64         print(f"Total bill: {bill}")
65     except ValueError as e:
66         print(f"Error: {e}")
```

```
PROBLEMS 10 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/souky/Downloads/#In above code billing logic must be re
Enter previous unit reading: 5000
Enter current unit reading: 6500
Enter customer type (residential/commercial/industrial): residential
Total bill amount: $1550.00
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/souky/Downloads/# In above code it should include multi
Enter previous unit reading: 5000
Enter current unit reading: 6500
Enter customer type (residential/commercial/industrial): residential
Total bill: 1648.5
PS C:\Users\souky>
```

Task 5: Final Bill Generation and Output Analysis

Scenario

The final electricity bill must present all values clearly.

Task Description

Develop the final Python application to:

- Calculate total bill:
- Total Bill = EC + FC + CC + ED
- Display:
 - o Energy Charges (EC)
 - o Fixed Charges (FC)
 - o Customer Charges (CC)
 - o Electricity Duty (ED)
 - o Total Bill Amount
- Analyze the program based on:
 - o Accuracy
 - o Readability
 - o Real-world applicability

Expected Output

- Complete electricity bill output
- Neatly formatted display
- Sample input/output
- Short analysis paragraph

```
C: > Users > souky > Downloads > #In above code it should print energy ch.py > calculate_customer_charge
1 #In above code it should print energy charges , fixed charges , customer charge ,electricity duty ,total bill amount.
2 def calculate_energy_charges(units_consumed,customer_type):
3     if customer_type == 'residential':
4         if units_consumed <= 100:
5             rate = 0.5
6         elif units_consumed <= 300:
7             rate = 0.75
8         else:
9             rate = 1.0
10    elif customer_type == 'commercial':
11        if units_consumed <= 100:
12            rate = 0.75
13        elif units_consumed <= 300:
14            rate = 1.0
15        else:
16            rate = 1.5
17    elif customer_type == 'industrial':
18        if units_consumed <= 100:
19            rate = 1.0
20        elif units_consumed <= 300:
21            rate = 1.5
22        else:
23            rate = 2.0
24    else:
25        raise ValueError("Invalid customer type")
26    return units_consumed * rate
27 def calculate_fixed_charges(customer_type):
28     if customer_type == 'residential':
29         return 50
30     elif customer_type == 'commercial':
31         return 100
```

```
... using f Untitled-3 • electric.py Untitled-6.py #In above code billing logic must be reu.py # In above code it should include multip.py • #In above code it should print energy ch.py
C > Users > souky > Downloads > #In above code it should print energy ch.py > calculate_bill
2 def calculate_energy_charges(units_consumed, customer_type):
24     else:
25         raise ValueError("Invalid customer type")
26     return units_consumed * rate
27 def calculate_fixed_charges(customer_type):
28     if customer_type == 'residential':
29         return 50
30     elif customer_type == 'commercial':
31         return 100
32     elif customer_type == 'industrial':
33         return 150
34     else:
35         raise ValueError("Invalid customer type")
36 def calculate_customer_charge(customer_type):
37     if customer_type == 'residential':
38         return 20
39     elif customer_type == 'commercial':
40         return 50
41     elif customer_type == 'industrial':
42         return 100
43     else:
44         raise ValueError("Invalid customer type")
45 def calculate_electricity_duty(bill_amount):
46     return bill_amount * 0.05 # 5% electricity duty
47 def calculate_bill(previous_units, current_units, customer_type):
48     units_consumed = current_units - previous_units
49     if units_consumed < 0:
50         raise ValueError("Current unit reading must be greater than or equal to previous unit reading")
51     energy_charges = calculate_energy_charges(units_consumed, customer_type)
52     fixed_charges = calculate_fixed_charges(customer_type)
53     customer_charge = calculate_customer_charge(customer_type)
```

```
... using f Untitled-3 • electric.py Untitled-6.py #In above code billing logic must be reu.py # In above code it should include multip.py • #In above code it should print energy ch.py
C > Users > souky > Downloads > #In above code it should print energy ch.py > calculate_bill
44         raise ValueError("Invalid customer type")
45 def calculate_electricity_duty(bill_amount):
46     return bill_amount * 0.05 # 5% electricity duty
47 def calculate_bill(previous_units, current_units, customer_type):
48     units_consumed = current_units - previous_units
49     if units_consumed < 0:
50         raise ValueError("Current unit reading must be greater than or equal to previous unit reading")
51     energy_charges = calculate_energy_charges(units_consumed, customer_type)
52     fixed_charges = calculate_fixed_charges(customer_type)
53     customer_charge = calculate_customer_charge(customer_type)
54     subtotal = energy_charges + fixed_charges + customer_charge
55     electricity_duty = calculate_electricity_duty(subtotal)
56     total_bill = subtotal + electricity_duty
57     return energy_charges, fixed_charges, customer_charge, electricity_duty, total_bill
58 if __name__ == "__main__":
59     try:
60         previous_units = int(input("Enter previous unit reading: "))
61         current_units = int(input("Enter current unit reading: "))
62         customer_type = input("Enter customer type (residential/commercial/industrial): ")
63         energy_charges, fixed_charges, customer_charge, electricity_duty, total_bill = calculate_bill(previous_units, current_units, customer_type)
64         print(f"Energy Charges: ${energy_charges:.2f}")
65         print(f"Fixed Charges: ${fixed_charges:.2f}")
66         print(f"Customer Charge: ${customer_charge:.2f}")
67         print(f"Electricity Duty: ${electricity_duty:.2f}")
68         print(f"Total Bill Amount: ${total_bill:.2f}")
69     except ValueError as e:
70         print(f"Error: {e}")
```

```
PROBLEMS 18 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-6
IndentationError: unindent does not match any outer indentation level
● PS C:\Users\souky> & C:/Users/souky/AppData/Local/Python/pythoncore-3.14-6
Enter previous unit reading: 5000
Enter current unit reading: 6000
Enter customer type (residential/commercial/industrial): residential
Energy Charges: $1000.00
Fixed Charges: $50.00
Customer Charge: $20.00
Electricity Duty: $53.50
Total Bill Amount: $1123.50
○ PS C:\Users\souky> 
```